



Social Network Analysis

Project 3

Objective

The student will apply different techniques to analyze the information in Social Networks.

Instructions

1. Map a Social Network using the reviewed API's or a Web Crawling/Scrapping technique.
2. Make a Visualization of the Social Network.
3. Make an analysis of the basic characteristics of the Network: Network Basic Characteristics, Degree Distribution, Global Clustering Coefficient and similar.
4. Make a Link Prediction Analysis identifying the most probable links to be appear in the network based on the different measures reviewed in class.
5. Make a report including:
 - a. Introduction. Describe the background of the mapped network.
 - b. Mapping Process. Describe the process to map the network.
 - c. Basic Characteristics and Visualization. Describe the basic characteristics analysis and include the visualization of the network.
 - d. Link Prediction. Describe the results of the link prediction analysis.

It is not sufficient to simply measure things, you need to discuss the insights you gained, always asking¹:

- What was your expectation?
- How do the results compare to your expectations?
- What did you learn from each quantity?

Homework Submission

Report

Submit the document of the report (.pdf).

Jupyter Notebook

Submit the Jupyter Notebook File (.ipynb).

Project will be presented during class on **August 5th, 2020**. Please, submit your work before the class, i.e. before 18:19 hrs.

¹ Network Science (Barabasi, 2018).



Grading Scheme

Rubric

	Excellent	Good	Fair	Needs Improvement
Introduction	10 pts	7.5 pts	5 pts	2.5 pts
Network Mapping	40 pts	35 pts	30 pts	20 pts
Network Visualization	20 pts	15 pts	10 pts	5 pts
Basic Measures	10 pts	7.5 pts	5 pts	2.5 pts
Link Prediction Analysis	20 pts	15 pts	10 pts	5 pts